

SELLER AND INVENTORY MANAGEMENT SYSTEM

DBMS – Task Name 3

1. Introduction

The Seller and Inventory Management System is a database application designed to maintain seller details, products, stock quantities, and inventory status. It establishes relationships between sellers, products, and inventory so that available and unavailable products can be tracked efficiently and inventory reports can be generated.

2. Objectives

- Create Seller, Product, and Inventory tables.
- Establish primary-key and foreign-key relationships.
- Maintain seller and seller-product information.
- Track available and unavailable products based on stock.
- Generate inventory status reports.

3. Database Design

Table	Important Fields	Purpose
Seller	seller_id (PK), seller_name, email, phone	Stores seller information
Product	product_id (PK), product_name, category, price, seller_id (FK)	Stores product and seller details
Inventory	inventory_id (PK), product_id (FK), quantity, status	Tracks stock and availability

4. SQL Table Creation

```
CREATE TABLE Seller (  
    seller_id INT PRIMARY KEY,  
    seller_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100),  
    phone VARCHAR(15)  
);  
  
CREATE TABLE Product (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(100) NOT NULL,  
    category VARCHAR(50),  
    price DECIMAL(10,2),  
    seller_id INT,  
    FOREIGN KEY (seller_id) REFERENCES Seller(seller_id)  
);  
  
CREATE TABLE Inventory (  
    inventory_id INT PRIMARY KEY,  
    product_id INT,  
    quantity INT NOT NULL,  
    status VARCHAR(20),  
    FOREIGN KEY (product_id) REFERENCES Product(product_id)  
);
```

5. Sample Data

Seller ID	Seller Name	Email	Phone
1	Arun Stores	arun@example.com	9876543210
2	Kavi Traders	kavi@example.com	9876501234

Product ID	Product	Category	Price	Seller ID
101	Laptop	Electronics	55000.00	1
102	Keyboard	Accessories	1200.00	1
103	Notebook	Stationery	80.00	2

Inventory ID	Product ID	Quantity	Status
1	101	12	Available
2	102	0	Unavailable
3	103	25	Available

6. Maintain Seller Product Information

The Product table connects each product to a seller through seller_id. This relationship makes it possible to identify which seller supplies each product. Product details can be inserted or updated using standard SQL commands.

```
UPDATE Product SET price = 56000.00 WHERE product_id = 101;
```

7. Track Available and Unavailable Products

A product is considered Available when its quantity is greater than zero. If the quantity is zero, the product is considered Unavailable. The following query displays the inventory status:

```
SELECT p.product_name, i.quantity,
CASE WHEN i.quantity > 0 THEN 'Available' ELSE 'Unavailable' END AS status
FROM Product p JOIN Inventory i ON p.product_id = i.product_id;
```

8. Inventory Status Report

The inventory report can combine seller, product, quantity, and status information. This helps management quickly identify available products and items that need restocking.

```
SELECT s.seller_name, p.product_name, i.quantity,
CASE WHEN i.quantity > 0 THEN 'Available' ELSE 'Unavailable' END AS status
FROM Seller s
JOIN Product p ON s.seller_id = p.seller_id
JOIN Inventory i ON p.product_id = i.product_id
ORDER BY s.seller_name, p.product_name;
```

9. Advantages

- Reduces manual inventory tracking.
- Maintains accurate seller and product information.
- Makes stock availability easy to identify.
- Supports quick inventory reporting.
- Uses primary and foreign keys to maintain data integrity.

10. Conclusion

The Seller and Inventory Management System provides a structured DBMS solution for managing sellers, products, and stock. Using related tables, keys, SQL queries, and inventory reports, the system can efficiently maintain product information and identify available or unavailable stock.